Quantitative Methods in Political Science: Introduction

Lab Week 1 – 10/11 September 2026

About me

Johann-Friedrich Salzmann (Joff)

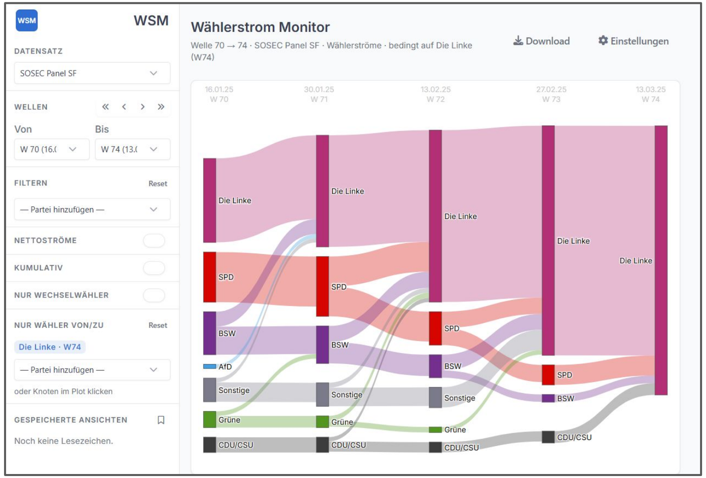
Research Interests

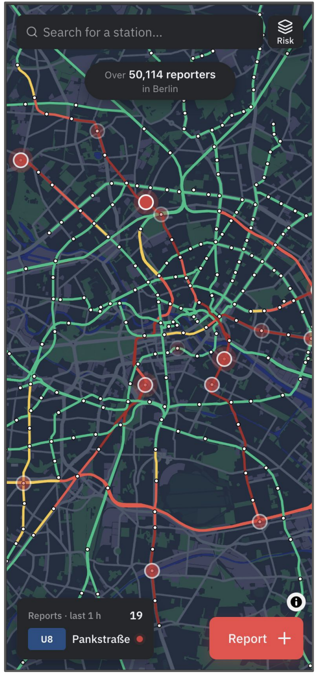
- Electoral Behaviour
- Vote Switching
- Opinion Dynamics

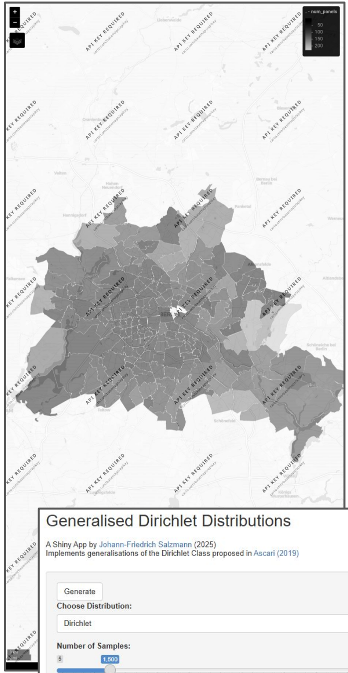
Methods Focus

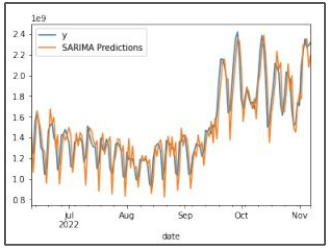
- Bayesian Statistics & Multi-Level Modelling
- Imputation of Categorical Data
- Compositional Data & Simplex Models

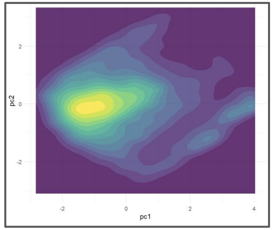
Some InspiRation

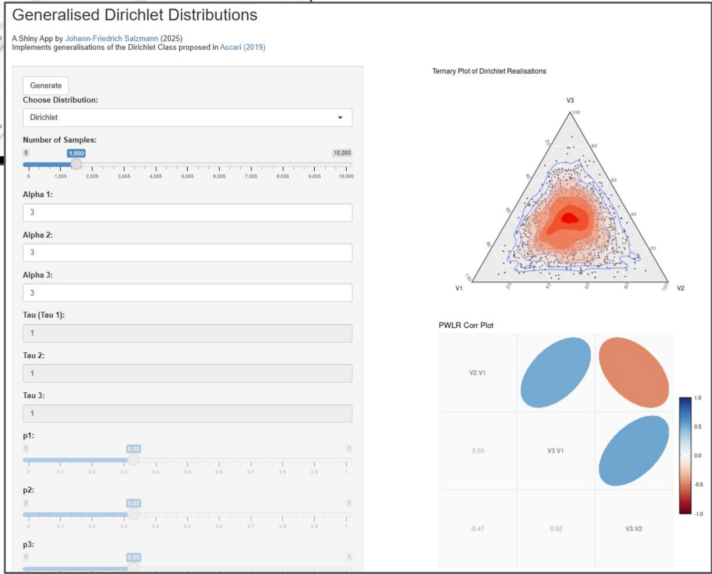


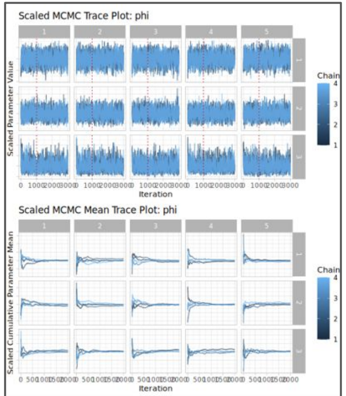


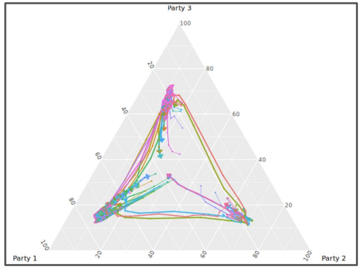












Who are we?

Joshua Rodrigues

- B.A. and M.A. Political Science at University of Mannheim
- PhD student at the CDSS
- Interests: Parliamentary rhetoric and party behaviour, text-as-data and statistical modelling

Who are we?

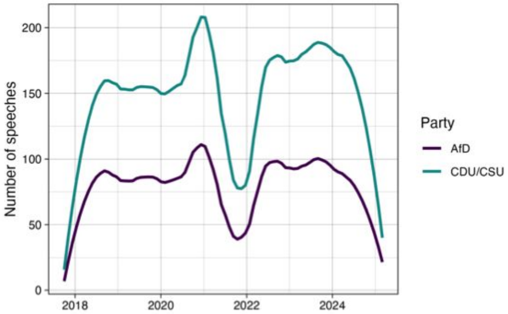


Figure 1: Number of Speeches delivered by the two parties over time. The plot shows monthly averages that are smoothed using the LOESS algorithm.

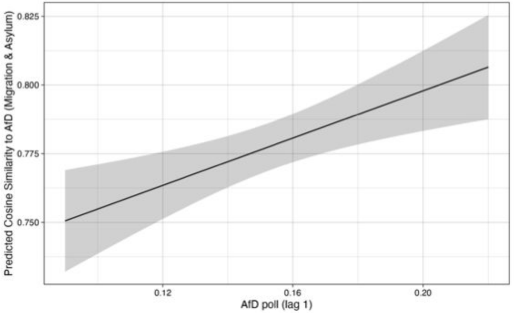


Figure 7: Expected cosine similarity of CDU/CSU speeches towards the AfD across different levels of AfD support on migration and asylum related speeches. Figure is based on the estimation results in table 3

Today’s plan

1. Get GitHub Org running
2. Get Positron & Quarto running
3. Test homework submission
4. Some very quick refresher on R (please see the Intro R course)
5. Centrality, summary stats, distributional plots
6. Some simple applications (*Try it*-examples)

Getting started with GitHub

- Open the GitHub organization (uni-mannheim-qm-2026)
- Create a homework repository for yourself via Github in the Browser:
  - “homework-” + your uni ID
  - Example: “homework-josalzma”, “homework-ab12cdef”
  - Keep it “private”, leave all other options untouched
- Don’t clone it yet! Refresh until you see some files in the repo.
- Only then, clone the repository
  - Green Code Button -> Open with Github Desktop

- Navigate to the ‘labs’ repository
- Clone the repository, too

Opening your repos in Positron / RStudio

- In Github Desktop
  - Windows: File -> Options -> Integrations -> External Editor: Custom
  - Mac: Github Desktop -> Settings -> Integrations -> External Editor: Custom
- Path: select your editor app:
  - Positron on Windows, it's in C:\Program Files\Positron, select Positron.exe
  - Positron on Mac, search your Apps, select Positron
  - Save, leave the options view
- Repository -> Open in external editor

- In Github Desktop, Repository -> Show in explorer/finder
  - Positron: Right-click .code-workspace file, Open with: Positron
  - RStudio: Double-click .Rproj file (should open RStudio)

- In Positron: Open Folder / File -> Open Workspace from File
- In RStudio: Open Project

Fill in & render the test homework

- In your editor open the homework repo via Workspace / Folder / R Project
- Open w00/w00.qmd and follow the instructions
- Push/Commit your complete test submission (qmd, html, pdf = typst)